Tuning Systems Exercises

Catharina Hock, Stefanie Hövermann

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Exponential Functions and Logarithms

Exercise 1: Powers

Calculate the following.

1. $2^5 =$ __________
2. $10^3 =$ __________
3. $5^0 =$ __________
4. $3^4 =$ __________
5. $2^{-2} =$ __________

Exercise 2: Exponential Growth

A bacteria culture starts with 200 bacteria. Every hour the number doubles.

1. Complete the table.

Time (h)	0	1	2	3	4
Bacteria	200	____	____	____	____

2. Write a formula for the number of bacteria after t hours.

3. How many bacteria are there after 6 hours?

Exercise 3: Exponential Decay

A radioactive substance loses half of its mass every day. Initially it has 80 g.

1. Complete the table.

Days	0	1	2	3	4
Mass (g)	80	____	____	____	____

2. Write a formula for the mass after t days.

3. After how many days is only 10 g left?

Exercise 4: Evaluating Logarithms

Calculate.

1. $\log_2(8) =$ ___________
2. $\log_2(32) =$ ___________
3. $\log_{10}(1000) =$ ___________
4. $\log_5(25) =$ ___________
5. $\log_3(81) =$ ___________

Exercise 5: Rewrite as Exponentials

Rewrite each logarithm as an exponential equation.

1. $\log_2(16) = 4$

2. $\log_5(125) = 3$

3. $\log_{10}(100) = 2$

Exercise 6: Rewrite as Logarithms

Rewrite each exponential equation as a logarithm.

1. $2^6 = 64$

2. $3^4 = 81$

3. $10^5 = 100000$

Exercise 7: Solve the Equations

1. $2^x = 32$

2. $10^x = 1000$

3. $3^x = 81$

4. $\log_2(x) = 5$

5. $\log_{10}(x) = 4$

Exercise 8: Word Problems

1. A savings account contains €500 and grows by 5% every year.
   (a) Write a formula for the balance after t years.
   (b) How much money is in the account after 4 years?

2. A rumor spreads through a school. The number of students who know it triples every day. On the first day, 4 students know the rumor.
   (a) Write an exponential model.
   (b) How many students know the rumor after 5 days?

Challenge

Without using a calculator, solve:

$$2^{x+1} = 64.$$

$$\log_2(x-1) = 3.$$

Explain in your own words why logarithms are called the *inverse* of exponential functions.

Exercise 9: The Natural Exponential Function

Recall that

$$\exp(x) = e^x.$$

Calculate the following. Give exact answers where possible.

1. $\exp(0) =$ ____________
2. $\exp(1) =$ ____________
3. $\exp(2) =$ ____________
4. $\exp(-1) \approx$ ____________

Rewrite each expression using powers of e.

1. $\exp(3)$

2. $\exp(-2)$

3. $\exp(x)$

Exercise 10: The Natural Logarithm

Recall that

$$\ln(x) = \log_e(x).$$

Calculate.

1. $\ln(e) =$ ____________
2. $\ln\left(e^3\right) =$ ____________
3. $\ln(1) =$ ____________
4. $\ln\left(e^{-2}\right) =$ ____________

Rewrite each equation.

1. $e^3 = 20.09\ldots$

2. $\ln(x) = 2$

3. $e^x = 7$

Exercise 11: Solving Equations

Solve the following.

1. $e^x = e^5$

2. $\ln(x) = 4$

3. $e^{2x} = e^8$

4. $\ln\big(e^{-5}\big)$

5. $e^{\ln(7)}$

6. $\ln\big(e^{12}\big)$